

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' -symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. Faulty RAM (Memory)	Computer does not boot, continuous beep sounds, random crashes, or blue screen errors.
2. Power Supply Unit (PSU) Failure	PC does not turn on, no lights or fans, or shuts down unexpectedly.
3. Hard Disk Drive (HDD) / SSD Failure	Slow performance, clicking noises (HDD), operating system fails to load, or "No boot device found" error.
4. Graphics Card (GPU) Problem	No display on the monitor, screen artifacts, distorted graphics, or display freezes.
5. CPU Overheating or Cooling Fan Failure	Computer becomes very hot, runs slowly, restarts automatically, or shuts down after a few minutes of use.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step	What to Check	Purpose
1. Check the Power Outlet	Plug another device (such as a phone charger or lamp) into the same outlet.	Verify that the wall outlet is supplying power.
2. Inspect the Charger and Cable	Look for cuts, frayed wires, bent connectors, or signs of overheating.	Ensure the charger and cable are not physically damaged.
3. Check the Charging Adapter LED (if available)	See whether the adapter's indicator light is on.	Confirm that the charger is receiving power.
4. Inspect the Laptop Charging Port	Check for dust, dirt, bent pins, or a loose charging socket. Clean carefully if needed.	Ensure the charger can make proper contact with the laptop.
5. Reconnect the Charger	Unplug and firmly reconnect both ends of the charger. Check if the charging LED turns on.	Rule out a loose connection.
6. Test with Another Compatible Charger	If available, use another charger with the correct voltage and wattage.	Determine whether the original charger is faulty.
7. Check the Battery (Removable Battery Only)	Remove the battery, clean the contacts, reconnect it, and try charging again.	Verify that the battery is properly connected.
8. Power On Without the Battery (If Removable)	Remove the battery and connect only the charger, then try turning on the laptop.	Check if the battery is defective while the laptop still works on AC power.
9. Observe Charging Indicators	Look for charging LEDs or battery icons that indicate charging status.	Determine whether the laptop detects the charger.

Step	What to Check	Purpose
10. Seek Hardware Repair if Needed	If none of the above steps work, the charging port, battery, or motherboard may be faulty.	Identify hardware issues that require professional repair.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

If you are using a virtual multimeter simulator (or need a sample observation for a lab report), you can use the following example:

Item Tested	Measured Voltage	Expected Range	Result
Standard USB Phone Charger (5V Output)	5.03 V DC	4.75 V – 5.25 V DC	Pass – Voltage is within the expected USB specification.

Procedure

1. Set the multimeter to DC Voltage (20V range or Auto Range).
2. Connect the black probe to the USB Ground (GND) pin.
3. Connect the red probe to the USB +5V (VBUS) pin.
4. Read the voltage displayed on the multimeter.
5. Compare the measured value with the standard USB range of 4.75V to 5.25V.

Observation

- Measured Voltage: 5.03 V DC
- Conclusion: The USB charger is operating normally because the measured voltage is within the acceptable USB voltage range.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?
Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the 12V Rail of an ATX Power Supply Using a Multimeter

Note: The following is a sample observation. If you perform the test yourself, record your actual measured voltage.

Procedure

1. Turn off the PC and disconnect it from the power outlet.
2. Disconnect the 24-pin ATX power connector from the motherboard.
3. Set the multimeter to DC Voltage (20V range or Auto Range).
4. Turn on the power supply (using a PSU tester or by safely bridging the PS_ON pin to ground if required).
5. Place the black multimeter probe on any black wire (Ground).
6. Place the red multimeter probe on a yellow wire (+12V).
7. Read the voltage displayed on the multimeter.
8. Compare the reading with the expected value of approximately 12V DC (acceptable range is about 11.4V–12.6V under the ATX specification).

Sample Measurement

Test Point	Measured Voltage	Expected Voltage	Result
Yellow (+12V) to Black (Ground)	12.08 V DC	Approximately 12 V DC	Pass – Voltage is normal.

If the Voltage Is Below 11.5V

A reading below 11.5V may indicate:

- A failing or weak Power Supply Unit (PSU).
- The PSU cannot provide stable power under load.
- Worn or damaged internal components (such as capacitors).
- This can cause random shutdowns, unexpected restarts, or system instability.

